

JAYRAJ M. PAMNANI

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SUMMARY

Software Engineer with experience building full-stack applications, backend services, cloud infrastructure, and ML systems. Skilled in developing scalable applications, REST APIs, CI/CD pipelines, and cloud-native services using Python, Java, TypeScript, React, Kubernetes, and AWS/GCP. Experienced in improving system performance, automating deployments, and delivering reliable production-ready solutions across software and ML environments.

PROFESSIONAL EXPERIENCE

Meesho - Software Engineer, Bengaluru, India

02/2023 - 08/2024

- Owned the development and production optimization of **real-time voice AI services**, initially supporting **voice-enabled search across 6 Indian languages and dialects** before transitioning to **conversational voice AI for customer support**.
- Designed and implemented **latency optimization improvements** across the voice pipeline by profiling end-to-end request paths, identifying network and service bottlenecks, deploying inference workloads across ~30 geographic regions, and **co-locating STT, LLM, TTS, and orchestration services** to reduce network round trips; helped reduce response latency from ~1.5s to **sub-second performance**.
- Owned **real-time responsiveness and turn-taking improvements** by implementing streaming across STT to LLM to TTS, tuning service routing, and designing **A/B tests for end-of-speech detection**, reducing the response wait window from 1–2s to approx. 400–500ms while improving conversational flow and reducing the robotic effect.

GBCS Group - Software Engineer, Calgary, Canada

11/2025 - 02/2026

- Implemented and optimized cloud infrastructure using Infrastructure as Code (IaC) to improve resource utilization, resulting in a **28% reduction in hosting and maintenance costs**.
- Improved CI/CD pipelines and deployment workflows, **accelerating release cycles by 40%** while maintaining 99.9% system availability.
- Helped redesign microservice boundaries and deployment workflows across a 5-engineer team, reducing technical debt, improving maintainability by 30%, and establishing patterns for future service development.
- Defined deployment standards for Docker/Kubernetes-based services, improving release reliability and reducing manual intervention.

New York University - Teaching Assistant – Machine Learning, New York, NY

09/2025 – 05/2026

- Guided **50+** graduate students through ML fundamentals, including preprocessing pipelining, supervised/unsupervised learning, Deep Learning, and model optimization techniques.
- Conducted weekly office hours to debug Python code, explained algorithms, taught ML topics, and assisted with PyTorch implementations.

Swaroop.ai - Software Engineering Intern, Ahmedabad, India

10/2022 - 01/2023

- Worked on voice AI systems for a creator platform, improving **speech-to-text and text-to-speech pipelines** for Indian languages, accents, and multilingual conversations, supporting approximately **10,000+ voice interactions per month**.
- Built and optimized speech-processing workflows including **audio preprocessing, noise reduction, transcription, model evaluation, and inference**, improving speech recognition accuracy by approximately **10–15%** and reducing average processing latency by **20%**.

SKILLS

Languages: Python, Java, Go, TypeScript, SQL, Bash | **Backend:** FastAPI, Django, Flask, Node.js, REST APIs, Microservices | **Frontend:** React, Next.js, Tailwind CSS, shadcn/ui | **Cloud & DevOps:** AWS, GCP, Azure, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD | **AWS Services:** EKS, Fargate, S3, SQS, RDS, CloudFront, ALB | **Databases & Data:** PostgreSQL, MongoDB, Snowflake, Kafka, Spark, Airflow | **ML/AI:** PyTorch, MLflow, Whisper, Coqui TTS, Model Serving

EDUCATION

New York University | Master's in Computer Engineering | GPA: 3.9/4.0 | May 2026

Relevant Coursework: High Performance Machine Learning, Machine Learning Operations, Deep Learning, Database Systems, Big Data

Parul University | Bachelor's in Computer Science & Engineering with Specialization in AI | GPA: 3.8/4.0 | May 2024

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Compiler Design, GPU computing, Pattern Recognition

PROJECTS

StreamForge (Distributed Video Transcoding & Secure Streaming Platform) | Node.js, AWS (Fargate, S3, SQS, CloudFront), FFmpeg

- Architected an event-driven, distributed video transcoding pipeline using AWS S3, SQS, and containerized FFmpeg workers on AWS Fargate, automatically processing raw video uploads into multi-resolution outputs (360p, 720p, 1080p) and adaptive HLS streams (.m3u8/.ts).
- Engineered a secure, low-latency media distribution layer backed by AWS CloudFront CDN and signed cookies with RSA key pairs, enforcing token-based authentication and session validation to restrict video playback to authorized users via protected shareable links.
- Developed the full-stack web platform and embeddable HLS video player with React, TypeScript, and Tailwind CSS featuring adaptive bitrate switching, provisioning reproducible cloud infrastructure with Terraform, Docker, and LocalStack for local development.

ActualBudget Transaction Categorizer | Python, FastAPI, PostgreSQL, MLflow, Docker, Kubernetes, Terraform

- Designed a reproducible ML platform around data ingestion, model evaluation, serving, experiment tracking, and GitOps-based infrastructure, enabling repeatable model updates and controlled deployment of categorization services.
- Built a Python-based data platform (API Backend) with PostgreSQL that ingests and categorizes financial transactions using a multi-stage ML classification pipeline.
- Engineered end-to-end data lifecycle management, implementing automated deployment pipelines (CI/CD) and Infrastructure as Code using Docker, Kubernetes, and Terraform for model serving and scaling.

HexDrop (Secure File Transferring WebApp) | Next.js, TypeScript, Prisma, PostgreSQL, AWS, Docker, K8s

- Developed a secure, full-stack file-sharing application using Next.js and TypeScript, implementing client-side encryption and managing the end-to-end user-facing flow.
- Designed a cloud-native deployment architecture on EKS with HPA, ALB Ingress, RDS PostgreSQL, External Secrets, and GitHub Actions to support secure, reproducible deployments and horizontal scaling.